

Maharashtra Administrative Data Management System

Project Overview

Build a full-stack web application to manage Maharashtra administrative and institutional data.

This platform will be used internally to manage hierarchical administrative records from State level down to Village/City/Town level along with Hospitals, Schools, Colleges, Population Records, Reports, Analytics, and Geographical Information.

The application must contain a Public Website and an Admin Dashboard.

Project Hierarchy

Maharashtra State

- Divisions
- Districts
 - Talukas
 - Villages
 - Cities
 - Towns

Each Village / City / Town may contain:

- Hospitals
 - Schools
 - Colleges
 - Population Records
-

Technology Stack

Frontend

- Next.js
- JavaScript
- Tailwind CSS
- Shadcn UI
- Zustand

- React Hook Form
- Zod
- TanStack Table
- Axios
- Recharts
- Leaflet
- React Leaflet
- Lucide React
- Sonner
- Framer Motion
- date-fns
- clsx
- tailwind-merge

Backend

- Next.js API Routes
- JWT Authentication
- bcryptjs
- Zod Validation
- XLSX
- Papa Parse
- jsPDF

Database

- PostgreSQL
- Prisma ORM

Deployment

- Vercel

Data Requirements

Create realistic sample data for:

- 1 Maharashtra State
- 6 Divisions
- 36 Districts
- 10–20 Talukas
- 20–50 Villages/Cities/Towns
- Hospitals
- Schools
- Colleges

- Population Records
-

Public Website

Navbar

Navigation Items:

- Explore
 - About
 - Contact
 - Login
-

Home Page (Explore)

Hero Section

Left Side

- Subtitle
- Title
- Description
- Global Search

Search should support:

- Division
- District
- Taluka
- Village
- City
- Town

Right Side

Statistics Cards

- Total Divisions
- Total Districts
- Total Talukas
- Total Cities
- Total Villages

- Total Towns
-

Division Section

Display Division Cards

Each Card Should Show:

- Division Name
 - District Count
 - Taluka Count
 - Village Count
 - City Count
 - Explore Button
-

Showcase Section

Left Side

- Image

Right Side

- Title
- Description
- CTA Buttons

Buttons:

- Explore Maharashtra
 - View Statistics
-

Footer

- Important Links
 - Quick Information
 - Copyright
-

About Page

Include:

- Platform Overview
 - Mission
 - Administrative Hierarchy
 - Institution Management
 - Population Tracking
 - Features Overview
-

Contact Page

Include:

- Contact Information
- Contact Form
- Leaflet Map
- FAQ Section

Contact Form Fields:

- Name
 - Email
 - Subject
 - Message
-

Login Page

Admin Only

Fields:

- Email
- Password

Features:

- Validation
- Show Password
- Remember Me
- JWT Login

Division Details Page

Hero Section

Breadcrumb:

Home > Division > Division Name

Left Side

- Division Label
- Division Name
- District Count
- Taluka Count
- Village Count
- City Count
- Town Count

Right Side

- Interactive Map

Below Hero

District Cards

Each Card:

- District Name
- Taluka Count
- Village Count
- City Count
- Open Button

District Details Page

Hero Section

Breadcrumb:

Home > Division > District

Left Side

- District Name
- Taluka Count
- Village Count
- City Count
- Town Count

Right Side

- District Map

Below Hero

Taluka Cards

Each Card:

- Taluka Name
- Village Count
- City Count
- Town Count
- Open Button

Taluka Details Page

Hero Section

Breadcrumb:

Home > Division > District > Taluka

Left Side

- Taluka Name
- Village Count
- City Count
- Town Count

Right Side

- Taluka Map

Below Hero

Category Cards

- Villages
- Cities
- Towns

Each Card Contains:

- Count
 - Explore Button
-

Village / City / Town Listing Page

Display Cards

Each Card Shows:

- Name
 - Population
 - Hospitals Count
 - Schools Count
 - Colleges Count
 - View Details
-

Village / City / Town Details Page

Hero Section

Breadcrumb Navigation

Left Side

- Place Type
- Place Name
- Address
- Map

Right Side

At A Glance Card

- Place Name
- Taluka Name

- District Name
 - Division Name
 - State Name
 - Population
 - Hospitals Count
 - Schools Count
 - Colleges Count
-

Details Section

General Information

- Name
- Type
- State
- Division
- District
- Taluka

Population Information

- Total Population
- Male Population
- Female Population
- Last Updated

Hospitals

Display Table

Schools

Display Table

Colleges

Display Table

Geographic Information

Leaflet Map

Related Places

Show Nearby Villages/Cities/Towns

Admin Dashboard

Layout

Left Side

- Sidebar

Top

- Navbar

Right Side

- Main Content Area
-

Sidebar Navigation

Dashboard

Administrative Management

- Divisions
- Districts
- Talukas
- Villages
- Cities
- Towns

Institution Management

- Hospitals
- Schools
- Colleges

Population

- Population Records

Reports & Analytics

- Reports

- Analytics

System

- Activity Logs
- Settings

User

- Profile
- Logout

Dashboard Home

Welcome Section

Show:

- Welcome Message
- Last Login
- Recent Updates

Statistics Cards

Administrative

- Total Divisions
- Total Districts
- Total Talukas
- Total Villages
- Total Cities
- Total Towns

Institutions

- Total Hospitals
- Total Schools
- Total Colleges

Population

- Total Population

Maharashtra Overview Map

Display Interactive Map

Analytics

Charts

- Population By Division
 - Population By District
 - Institution Distribution
 - Growth Statistics
-

Recent Activity

Timeline View

Quick Actions

- Add Division
 - Add District
 - Add Taluka
 - Add Village
 - Add Hospital
 - Add School
 - Add College
-

Administrative Modules

Create Full CRUD For:

- Divisions
- Districts
- Talukas
- Villages
- Cities
- Towns

Features:

- Create
 - Update
 - Delete
 - Search
 - Filters
 - Pagination
 - Sorting
-

Institution Modules

Create Full CRUD For:

- Hospitals
- Schools
- Colleges

Features:

- Image Upload
 - Search
 - Filters
 - Pagination
-

Population Module

Manage:

- Population
- Male Population
- Female Population
- Last Updated

Analytics Included

Reports Module

Generate:

- Division Reports

- District Reports
- Taluka Reports
- Hospital Reports
- School Reports
- College Reports
- Population Reports

Export Formats:

- PDF
 - CSV
 - Excel
-

Analytics Module

Charts:

- Population Trends
 - Top Districts
 - Institution Density
 - Administrative Distribution
-

Activity Logs

Track:

- Create
- Update
- Delete
- Login
- Export
- Import

Display:

- User
 - Action
 - Entity
 - Date
 - Time
-

Settings

Profile

- Name
- Email
- Avatar

Security

- Change Password

Preferences

- Theme
 - Dashboard Settings
-

Required Features

- JWT Authentication
 - Protected Routes
 - Responsive Design
 - Dark Mode
 - Search
 - Filters
 - Pagination
 - Loading States
 - Skeleton Loaders
 - Error Handling
 - Empty States
 - Toast Notifications
 - Form Validation
 - File Upload
 - Dynamic Routing
 - Breadcrumb Navigation
 - Reusable Components
 - API Documentation
-

Submission Requirements

Mandatory:

1. GitHub Repository
2. README.md
3. Database Schema
4. API Documentation

Optional:

1. Demo Video (2–5 Minutes)
2. Postman Collection

Deadline

5 Days

AI Usage Policy

AI tools are allowed.

However, the developer must be able to explain:

- Project Architecture
- Database Design
- API Design
- Authentication Flow
- Folder Structure
- Key Technical Decisions

during review.